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Sequences and Limits: Solutions

Exercises 2.2.2, 2.3.1–2.3.4, 2.4.3 and Theorem 2.3.3

Throughout, we use **Definition 2.2.3**: a sequence (a_n) converges to a if for every $\varepsilon > 0$ there exists $N \in \mathbf{N}$ such that $n \geq N$ implies $|a_n - a| < \varepsilon$. We freely use the Archimedean Property: for every real t there is $N \in \mathbf{N}$ with $N > t$.

Exercise 2.2.2

(a) $\lim \dfrac{2n+1}{5n+4} = \dfrac{2}{5}$.

Proof. For every $n \in \mathbf{N}$,

$$\left|\frac{2n+1}{5n+4} - \frac{2}{5}\right| = \left|\frac{5(2n+1) - 2(5n+4)}{5(5n+4)}\right| = \frac{3}{25n+20} < \frac{3}{25n}.$$

Let $\varepsilon > 0$ and choose $N \in \mathbf{N}$ with $N > \dfrac{3}{25\varepsilon}$. Then for $n \geq N$ we have $\dfrac{3}{25n} \leq \dfrac{3}{25N} < \varepsilon$, so $\left|\frac{2n+1}{5n+4} - \frac{2}{5}\right| < \varepsilon$. □

(b) $\lim \dfrac{2n^2}{n^3+3} = 0$.

Proof. For every $n \in \mathbf{N}$,

$$\left|\frac{2n^2}{n^3+3} - 0\right| = \frac{2n^2}{n^3+3} < \frac{2n^2}{n^3} = \frac{2}{n}.$$

Let $\varepsilon > 0$ and choose $N \in \mathbf{N}$ with $N > 2/\varepsilon$. For $n \geq N$, $\frac{2}{n} \leq \frac{2}{N} < \varepsilon$, hence $\left|\frac{2n^2}{n^3+3}\right| < \varepsilon$. □

(c) $\lim \dfrac{\sin(n^2)}{\sqrt[3]{n}} = 0$.

Proof. Since $|\sin x| \leq 1$ for all real x,

$$\left|\frac{\sin(n^2)}{\sqrt[3]{n}} - 0\right| \leq \frac{1}{\sqrt[3]{n}}.$$

Let $\varepsilon > 0$ and choose $N \in \mathbf{N}$ with $N > 1/\varepsilon^3$. For $n \geq N$ we have $n > 1/\varepsilon^3$, so $\sqrt[3]{n} > 1/\varepsilon$, i.e. $\frac{1}{\sqrt[3]{n}} < \varepsilon$. Therefore $\left|\frac{\sin(n^2)}{\sqrt[3]{n}}\right| < \varepsilon$ for all $n \geq N$. □

Theorem 2.3.3 (Algebraic Limit Theorem)

Let $\lim a_n = a$ and $\lim b_n = b$. We first need a lemma.

Lemma (Convergent sequences are bounded)**.** If $(x_n) \to x$, then there is $M > 0$ with $|x_n| \leq M$ for all $n \in \mathbf{N}$.

Proof. With $\varepsilon = 1$, there is N such that $|x_n - x| < 1$ for $n \geq N$, so $|x_n| < |x| + 1$ for $n \geq N$. Put $M = \max\{|x_1|, \ldots, |x_{N-1}|, |x| + 1\}$. Then $|x_n| \leq M$ for all n. □

(i) $\lim(ca_n) = ca$.

Proof. If $c = 0$, then (ca_n) is the constant sequence 0, which converges to $0 = ca$. Suppose $c \neq 0$ and let $\varepsilon > 0$. Choose N so that $|a_n - a| < \varepsilon/|c|$ for $n \geq N$. Then for $n \geq N$,

$$|ca_n - ca| = |c|\,|a_n - a| < |c| \cdot \frac{\varepsilon}{|c|} = \varepsilon.$$

□

(ii) $\lim(a_n + b_n) = a + b$.

Proof. Let $\varepsilon > 0$. Choose N_1 with $|a_n - a| < \varepsilon/2$ for $n \geq N_1$ and N_2 with $|b_n - b| < \varepsilon/2$ for $n \geq N_2$. Let $N = \max\{N_1, N_2\}$. For $n \geq N$, by the triangle inequality,

$$|(a_n + b_n) - (a + b)| \leq |a_n - a| + |b_n - b| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

□

(iii) $\lim(a_n b_n) = ab$.

Proof. By the lemma there is $M > 0$ with $|b_n| \leq M$ for all n. Adding and subtracting ab_n,

$$|a_n b_n - ab| = |a_n b_n - ab_n + ab_n - ab| \leq |b_n|\,|a_n - a| + |a|\,|b_n - b| \leq M|a_n - a| + |a|\,|b_n - b|.$$

Let $\varepsilon > 0$. Choose N_1 with $|a_n - a| < \dfrac{\varepsilon}{2M}$ for $n \geq N_1$, and N_2 with $|b_n - b| < \dfrac{\varepsilon}{2(|a| + 1)}$ for $n \geq N_2$. For $n \geq N = \max\{N_1, N_2\}$,

$$|a_n b_n - ab| < M \cdot \frac{\varepsilon}{2M} + |a| \cdot \frac{\varepsilon}{2(|a| + 1)} < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

□

(iv) $\lim(a_n/b_n) = a/b$, provided $b \neq 0$.

Proof. By (iii) it suffices to show $(1/b_n) \to 1/b$, since $a_n/b_n = a_n \cdot (1/b_n)$.

Taking $\varepsilon_0 = |b|/2 > 0$, there is N_1 with $|b_n - b| < |b|/2$ for $n \geq N_1$; then $|b_n| \geq |b| - |b - b_n| > |b|/2$. In particular $b_n \neq 0$ for $n \geq N_1$, and

$$\left|\frac{1}{b_n} - \frac{1}{b}\right| = \frac{|b - b_n|}{|b|\,|b_n|} < \frac{2}{|b|^2}\,|b_n - b| \qquad (n \geq N_1).$$

Let $\varepsilon > 0$ and choose N_2 with $|b_n - b| < \dfrac{\varepsilon |b|^2}{2}$ for $n \geq N_2$. For $n \geq \max\{N_1, N_2\}$,

$$\left|\frac{1}{b_n} - \frac{1}{b}\right| < \frac{2}{|b|^2} \cdot \frac{\varepsilon |b|^2}{2} = \varepsilon.$$

So $(1/b_n) \to 1/b$, and by (iii) $(a_n/b_n) \to a \cdot \frac{1}{b} = \frac{a}{b}$. □

Exercise 2.3.1

Let $x_n \geq 0$ for all $n \in \mathbf{N}$.

(a) If $(x_n) \to 0$, then $(\sqrt{x_n}) \to 0$.

Proof. Let $\varepsilon > 0$. Since $\varepsilon^2 > 0$, there is N such that $|x_n - 0| = x_n < \varepsilon^2$ for all $n \geq N$. Because the square root function is increasing on $[0, \infty)$, for $n \geq N$,

$$|\sqrt{x_n} - 0| = \sqrt{x_n} < \sqrt{\varepsilon^2} = \varepsilon. \qquad \square$$

(b) If $(x_n) \to x$, then $(\sqrt{x_n}) \to \sqrt{x}$.

Proof. First, $x \geq 0$, so $\sqrt{x}$ makes sense. Indeed, if $x < 0$, take $\varepsilon = |x|$; then for large n, $x_n < x + |x| = 0$, contradicting $x_n \geq 0$.

Case $x = 0$: this is part (a).

Case $x > 0$: For every n,

$$|\sqrt{x_n} - \sqrt{x}| = \frac{|\sqrt{x_n} - \sqrt{x}|\,(\sqrt{x_n} + \sqrt{x})}{\sqrt{x_n} + \sqrt{x}} = \frac{|x_n - x|}{\sqrt{x_n} + \sqrt{x}} \leq \frac{|x_n - x|}{\sqrt{x}},$$

using $\sqrt{x_n} \geq 0$. Let $\varepsilon > 0$ and choose N with $|x_n - x| < \varepsilon\sqrt{x}$ for $n \geq N$. Then for $n \geq N$,

$$|\sqrt{x_n} - \sqrt{x}| < \frac{\varepsilon\sqrt{x}}{\sqrt{x}} = \varepsilon. \qquad \square$$

Exercise 2.3.2

Assume $(x_n) \to 2$; we use only Definition 2.2.3.

(a) $\left(\dfrac{2x_n - 1}{3}\right) \to 1$.

Proof. For every n,

$$\left|\frac{2x_n - 1}{3} - 1\right| = \left|\frac{2x_n - 4}{3}\right| = \frac{2}{3}|x_n - 2|.$$

Let $\varepsilon > 0$ and choose N with $|x_n - 2| < \frac{3\varepsilon}{2}$ for $n \geq N$. Then for $n \geq N$, $\left|\frac{2x_n - 1}{3} - 1\right| < \frac{2}{3} \cdot \frac{3\varepsilon}{2} = \varepsilon$. $\square$

(b) $(1/x_n) \to 1/2$.

Proof. For $x_n \neq 0$,

$$\left|\frac{1}{x_n} - \frac{1}{2}\right| = \frac{|2 - x_n|}{2|x_n|}.$$

Using the definition with $\varepsilon_0 = 1$, there is N_1 such that $|x_n - 2| < 1$, hence $1 < x_n < 3$, for all $n \geq N_1$. So for $n \geq N_1$, $|x_n| > 1$ and

$$\left|\frac{1}{x_n} - \frac{1}{2}\right| < \frac{|x_n - 2|}{2}.$$

Let $\varepsilon > 0$ and choose N_2 with $|x_n - 2| < 2\varepsilon$ for $n \geq N_2$. For $n \geq N = \max\{N_1, N_2\}$,

$$\left|\frac{1}{x_n} - \frac{1}{2}\right| < \frac{2\varepsilon}{2} = \varepsilon. \qquad \square$$

Exercise 2.3.3 (Squeeze Theorem)

If $x_n \le y_n \le z_n$ for all $n \in \mathbf{N}$ and $\lim x_n = \lim z_n = l$, then $\lim y_n = l$.

Proof. Let $\varepsilon > 0$. Choose N_1 with $|x_n - l| < \varepsilon$ for $n \ge N_1$; in particular $l - \varepsilon < x_n$. Choose N_2 with $|z_n - l| < \varepsilon$ for $n \ge N_2$; in particular $z_n < l + \varepsilon$. For $n \ge N = \max\{N_1, N_2\}$,

$$l - \varepsilon < x_n \le y_n \le z_n < l + \varepsilon,$$

that is, $|y_n - l| < \varepsilon$. Hence $(y_n) \to l$. □

Exercise 2.3.4

The statement in the image is cut off. The solutions below are for the standard version of this exercise (Abbott, Understanding Analysis*, 2nd ed.):* Let $(a_n) \to 0$ and use the Algebraic Limit Theorem to compute each limit, assuming the fractions are always defined.

(a) $\lim \dfrac{1 + 2a_n}{1 + 3a_n - 4a_n^2}$.

By ALT (i)–(iii), $1 + 2a_n \to 1 + 2 \cdot 0 = 1$ and $1 + 3a_n - 4a_n^2 \to 1 + 0 - 4 \cdot 0 \cdot 0 = 1 \neq 0$. By ALT (iv) the limit is $\dfrac{1}{1} = \boxed{1}$.

(b) $\lim \dfrac{(a_n + 2)^2 - 4}{a_n}$.

Since the fraction is defined, $a_n \neq 0$, and $\dfrac{(a_n + 2)^2 - 4}{a_n} = \dfrac{a_n^2 + 4a_n}{a_n} = a_n + 4$. By ALT (ii) this converges to $0 + 4 = \boxed{4}$.

(c) $\lim \dfrac{\frac{2}{a_n} + 3}{\frac{3}{a_n} + 5}$.

Multiplying numerator and denominator by $a_n \neq 0$ gives $\dfrac{2 + 3a_n}{3 + 5a_n}$. By ALT, $2 + 3a_n \to 2$ and $3 + 5a_n \to 3 \neq 0$, so by (iv) the limit is $\boxed{\frac{2}{3}}$.

Exercise 2.4.3

(a) The sequence $\sqrt{2},\ \sqrt{2 + \sqrt{2}},\ \sqrt{2 + \sqrt{2 + \sqrt{2}}}, \ldots$ converges, and its limit is 2.

Proof. The sequence is defined recursively by $x_1 = \sqrt{2}$ and $x_{n+1} = \sqrt{2 + x_n}$. Clearly $x_n > 0$ for all n.

Bounded above by 2. $x_1 = \sqrt{2} < 2$. If $x_n < 2$, then $x_{n+1} = \sqrt{2 + x_n} < \sqrt{4} = 2$. By induction, $0 < x_n < 2$ for all n.

Increasing. Since both sides are positive,

$$x_{n+1} > x_n \iff 2 + x_n > x_n^2 \iff (2 - x_n)(1 + x_n) > 0,$$

which holds because $0 < x_n < 2$. So (x_n) is increasing.

By the Monotone Convergence Theorem, (x_n) converges; let $L = \lim x_n$. Since (x_{n+1}) is a tail of the same sequence, $x_{n+1} \to L$ as well. Squaring the recursion gives $x_{n+1}^2 = 2 + x_n$, and by the Algebraic Limit Theorem,

$$L^2 = 2 + L \quad \Longrightarrow \quad (L-2)(L+1) = 0 \quad \Longrightarrow \quad L = 2 \text{ or } L = -1.$$

Since the sequence is increasing, $x_n \geq x_1 = \sqrt{2}$ for all n, so $L \geq \sqrt{2} > 0$ (Order Limit Theorem). Hence $\lim x_n = 2$. □

(b) The sequence $\sqrt{2},\ \sqrt{2\sqrt{2}},\ \sqrt{2\sqrt{2\sqrt{2}}}, \ldots$ converges, and its limit is 2.

Proof. Here $y_1 = \sqrt{2}$ and $y_{n+1} = \sqrt{2y_n}$, and $y_n > 0$ for all n.

Bounded above by 2. $y_1 < 2$, and if $y_n < 2$ then $y_{n+1} = \sqrt{2y_n} < \sqrt{4} = 2$. By induction $0 < y_n < 2$ for all n.

Increasing. Since both sides are positive,

$$y_{n+1} > y_n \iff 2y_n > y_n^2 \iff y_n(2 - y_n) > 0,$$

which holds since $0 < y_n < 2$.

By the Monotone Convergence Theorem, (y_n) converges to some L, and $y_{n+1} \to L$ too. From $y_{n+1}^2 = 2y_n$ and the Algebraic Limit Theorem, $L^2 = 2L$, so $L = 0$ or $L = 2$. Since $y_n \geq y_1 = \sqrt{2}$ for all n, $L \geq \sqrt{2}$, so $L = 2$. □

Remark (alternative for (b)). By induction, $y_n = 2^{\,1-2^{-n}}$: indeed $y_1 = 2^{1/2}$, and if $y_n = 2^{1-2^{-n}}$ then $y_{n+1} = (2 \cdot 2^{1-2^{-n}})^{1/2} = 2^{1-2^{-(n+1)}}$. Since $2^{-n} \to 0$ and $t \mapsto 2^t$ is continuous, $y_n \to 2^1 = 2$, confirming the answer.